

Dynamic Labor Supply and Lifecycle Responses

Labor I — Week 3

Teaching deck draft

Central question and motivation

- How do labor-supply choices change once workers can shift work across time instead of choosing hours one period at a time?
- Week 2's static condition survives, but now assets, anticipation, and lagged choices matter for what an estimate means.
- Dynamic evidence is useful because temporary shocks, permanent shocks, and lifecycle changes do not identify the same object.

Dynamic labor-supply problem

$$\max_{\{c_t, h_t, a_{t+1}\}_{t=0}^T} \sum_{t=0}^T \beta^t u(c_t, \bar{h} - h_t) \quad \text{s.t.} \quad a_{t+1} = (1+r)a_t + y_t + w_t h_t - T_t(w_t h_t, y_t) - c_t$$

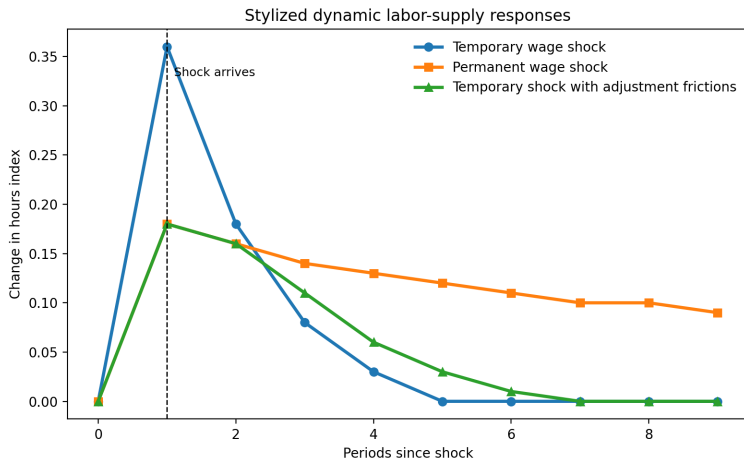
- Current work changes both current resources and future feasible choices.
- The shadow value of wealth is therefore an intertemporal state variable.

Intratemporal condition and Frisch interpretation

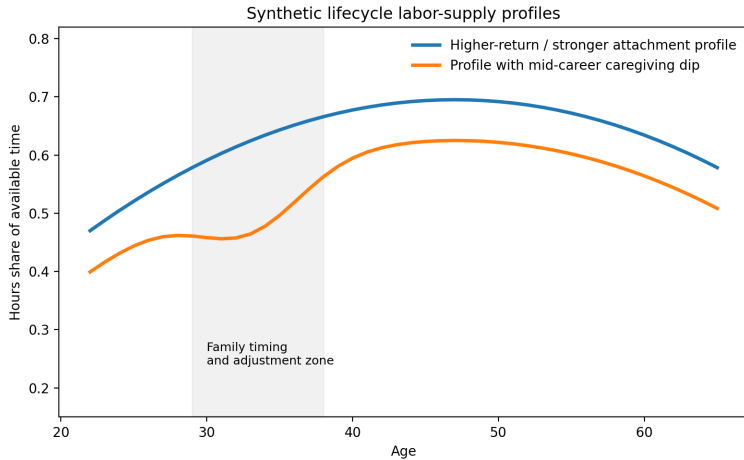
$$\frac{u_{\ell,t}(c_t, \ell_t)}{u_{c,t}(c_t, \ell_t)} = w_t(1 - \tau_t^m), \quad \ell_t = \bar{h} - h_t$$

- The Frisch elasticity holds fixed the marginal utility of wealth.
- That makes it an intertemporal substitution object, not a generic policy elasticity.
- Permanent wage changes mix substitution and wealth effects, so they are not clean Frisch experiments.

Temporary versus permanent wage changes



Why lifecycle profiles matter



- Hours profiles over age are equilibrium objects shaped by wages, family timing, health, and retirement.
- Human-capital accumulation means current hours can raise future wages.
- That is why Week 3 naturally feeds Week 4: labor supply and skill accumulation are already intertwined.

Empirical design map

Design	Main variation	Object learned
Randomized temporary wage variation	Short-run exogenous pay shift	Intertemporal substitution / Frisch-like response
Panel wage-hours variation	Within-worker changes over time	Dynamic reduced-form response
Tax timing or reform design	Timing or slope of net-of-tax schedule	Horizon-specific labor-supply response
Lifecycle structural model	Joint restrictions on wages, hours, savings, family states	Parameters for richer counterfactual policy analysis

Persistence and adjustment costs

$$\tilde{u}_t = u(c_t, \bar{h} - h_t) - \Phi(h_t - h_{t-1}), \quad \Phi(h_t - h_{t-1}) = \frac{\kappa}{2}(h_t - h_{t-1})^2$$

- Small observed short-run responses need not imply small desired reoptimization.
- Adjustment costs, participation costs, and rigid schedules can smooth the measured response.
- This is the bridge from micro estimates to the broader elasticity debate.

What the Week 3 lab identifies

- Reproduce: a bounded response to randomized temporary wage variation in the spirit of Fehr and Goette.
- Diagnose: ask whether the response is best read as intertemporal substitution or a broader labor-supply object.
- Transfer: compare the same temporary incentive under low and high adjustment-cost environments.

- If current work raises future wages, labor supply already contains human-capital investment.
- Lifecycle differences in hours can therefore reflect changing returns to experience, not only changing tastes.
- Week 4 makes that implicit investment margin explicit.

Takeaways

- Dynamic labor supply is about timing, not just levels.
- Temporary and permanent shocks identify different objects.
- The Frisch elasticity is conditional on the intertemporal shadow value of wealth.
- Persistence and adjustment frictions are part of the estimand, not a nuisance afterthought.